Logitech Sales Forecasting Report

MAS 640 — Time Series Analysis & Forecasting | Individual Project

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Overview

This project develops an automated, multi-model forecasting framework for Logitech monthly product sales data. The dataset spans **44 months** across 3 regions and 11 product groups, organized in a three-level category hierarchy (Region → Product Group → Sub-category). The raw data was intentionally messy—a realistic industry scenario—requiring a thorough cleaning pipeline before any modeling.

After cleaning, **87 viable product series** were identified and independently forecasted using a **14-model comparison framework** spanning four families: classical statistical models (SARIMAX, Log-SARIMAX, ETS), 7 automated Nixtla StatsForecast models, 2 machine learning models (Gradient Boosting, Random Forest), and 2 baselines (Seasonal Naïve, Drift). Each series selects its own best model through expanding-window cross-validation, and a diversity-based ensemble provides a safety net. The framework achieved **MAPE below 10% on 62% of series** and below 20% on 87%, delivering reliable 12-month forecasts across the full portfolio.

Data Loading & Cleaning

The raw dataset is *intentionally messy*—a common pattern in interview use-case scenarios and real-world sales data. Eight distinct quality issues were identified, each addressed with a documented rationale:

Issue	Solution
Empty separator rows	Remove rows where all values are NaN
Wide format (1 col/month)	Melt to long format (date, sales)
Date format `YY-Mon`	Parse to proper `datetime`
Leading zeros	Trim pre-launch periods
Trailing zeros ($\geq$30)	Flag as discontinued, remove
Negative values	Replace with 0 (data entry errors)
Duplicate category keys	Keep as individual series
NULL in Category3	Retain, label as `NULL`

Key considerations:
Duplicate category keys are not actual duplicates—they represent distinct sub-products sharing the same hierarchy (e.g., `B_C_W` has series of 350M, 310M, and 632 units). Aggregating them would cause signal contamination: the 632-unit series has completely different dynamics than the 350M one.
After cleaning: series with average monthly sales <1,000 were removed as noise (test SKUs or data artifacts), yielding **87 viable series** with $\geq$36 months of history across **3 regions** and **11 product groups**.

Exploratory Data Analysis (EDA)

Category Hierarchy & Regional Analysis

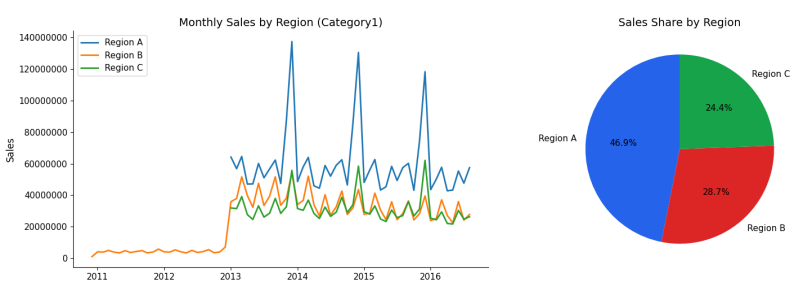

Figure 1: Monthly sales by region and share. Region A leads at 46.9%.

Sales are distributed across three regions with Region A dominating at 46.9%, followed by B (28.7%) and C (24.4%). The time series shows strong cyclical behavior with pronounced December spikes across all regions, along with a general upward trend that flattens post-2013.
The product hierarchy follows a clear **Pareto distribution**: groups C (30.7%), A (19.4%), I (14.1%), and G (11.8%) account for 76% of total sales.

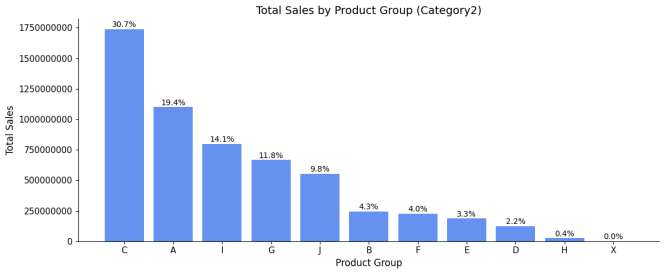

Figure 2: Product group Pareto: top 4 groups drive 76% of volume.

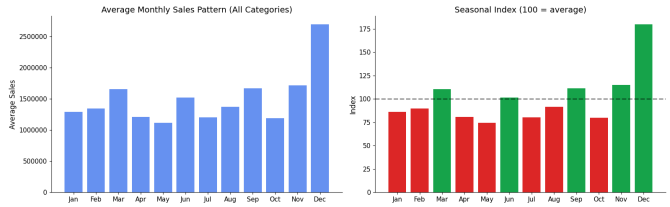

Figure 3: Seasonal pattern: Dec index 183, May trough at 75.

Time Series Patterns & Seasonality

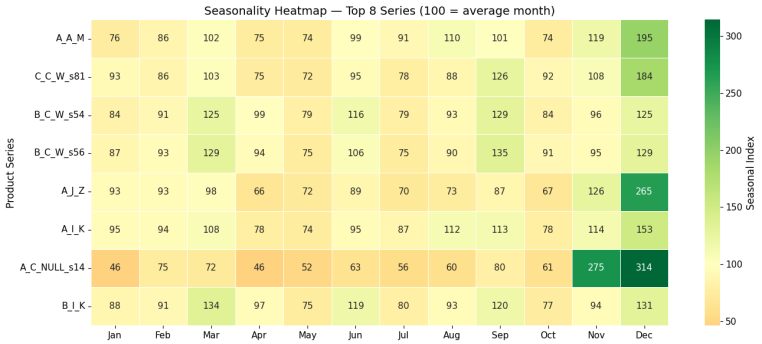

Figure 4: Seasonality heatmap — top 8 series. December exceeds 300 for `A_C_NULL_s14`.

Seasonality is the dominant feature: December indices range from 125 to 314, driven by holiday shopping. The Oct–Nov buildup is consistent, while May represents the annual trough.

YoY growth shows most top series declining in 2015–2016, suggesting market maturation. Only a few series (e.g., `A_I_K`) maintained positive growth.

Correlation analysis reveals that many top series co-move (correlations 0.7–0.9), suggesting shared macro drivers like economic conditions and seasonal shopping patterns.

Time Series Diagnostics

Before forecasting, we validated the statistical properties of each series through three diagnostic tests:

Stationarity Tests (ADF & KPSS): Most series are non-stationary in levels but become stationary after first differencing. This is expected for trending sales data. Models handle non-stationarity internally—SARIMAX via the d and D differencing parameters, ETS via trend components. We test stationarity to *understand* the data, not to manually difference before modeling.

Seasonal Decomposition: Additive decomposition isolates trend, seasonal, and residual components for each series. Strong seasonal patterns confirmed across all major series, with residuals showing no systematic structure—indicating the decomposition captures the signal well.

ACF/PACF Analysis: Significant autocorrelation at lags 1, 12, and 24 across most series confirms monthly and annual dependencies, justifying the use of ARIMA-class models with seasonal components. The gradual ACF decay pattern is characteristic of non-stationary processes requiring differencing, while the PACF cutoffs help determine AR order selection for SARIMAX.

These diagnostics confirmed that the data exhibits the classic characteristics of monthly sales series: **trend + strong seasonality + autocorrelation**—making it well-suited for the SARIMAX, ETS, and Nixtla model families selected.

Forecasting All Product Series

Framework & Cross-Validation

We apply a **14-model comparison** across 4 families. The key design decision is **per-series model selection**: rather than forcing one model on all 87 series, each series independently selects its best model through cross-validated RMSE. This ensures the forecast adapts to each series' unique dynamics.

The **expanding-window cross-validation** (Figure 5) uses 2 folds with 12-month validation windows. This maximizes training data in each fold while providing honest out-of-sample evaluation. Both **RMSE** (for within-series optimization) and **MAPE** (for cross-series comparison) are computed. The final model is retrained on all 44 months before producing the 12-month forecast.

For SARIMAX, we use **AIC** instead of BIC for model selection. With $n = 44$, BIC's penalty ($\log n \approx 3.8$ per parameter vs. AIC's fixed 2) aggressively strips seasonal components. AIC is asymptotically equivalent to leave-one-out CV error and is the standard recommendation for forecasting (Hyndman & Athanasopoulos).

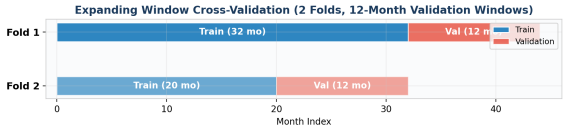

Figure 5: Expanding-window CV: Fold 1 trains 32 mo, Fold 2 trains 20 mo; both validate 12 mo.

Family	Models
Statistical	SARIMAX, Log-SARIMAX, ETS (10 configs)
Nixtla (7)	AutoARIMA, AutoETS, AutoTheta, AutoCES, DynOptTheta, OptTheta, MSTL
ML	Gradient Boosting, Random Forest
Baselines	Seasonal Naïve, Drift
Ensemble	Mean of best per family

Results: Model Performance

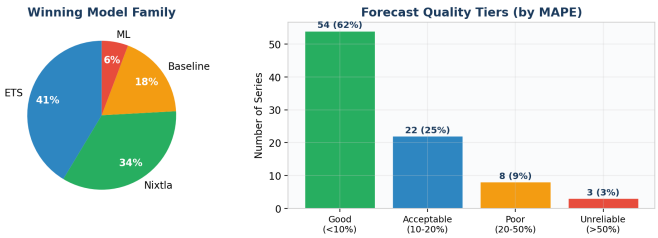

Figure 6: Left: winning model family distribution. Right: forecast quality tiers by MAPE.

ETS and Nixtla dominate, together winning **75%** of series. This validates including both classical exponential smoothing and modern automated approaches—different series respond better to different model families.

ML underperforms (6%) due to insufficient training data (∼20 usable rows after feature engineering). With only ∼3.5 seasonal cycles, tree-based models cannot learn robust lag patterns.

Baselines win 18% of series—these are genuinely noisy series where added model complexity actually *hurts* forecast accuracy by overfitting to noise.

Quality: 62% of series achieve MAPE <10% (Good), and 87% are below 20%. Only 3 series (3%) are unreliable (>50% MAPE).

Top Performing Models & Metrics

Series	Best Model	Type	RMSE	MAPE	Series	Best Model	Type	RMSE	MAPE
A_A_M	Drift_50pct	Base	1.04M	3.9%	A_J_Z	ETS(M,Md,M)	ETS	373K	5.7%
B_C_W_s54	ETS(M,A,M)	ETS	573K	8.0%	A_I_K	SeasonalNaive	Base	186K	2.3%
C_C_W_s79	Nixtla_AutoARIMA	Nixtla	168K	2.4%	C_G_F	Nixtla_AutoETS	Nixtla	140K	3.0%
A_D_NULL	Nixtla_OptTheta	Nixtla	89K	3.1%	B_A_N	SeasonalNaive	Base	89K	2.6%
B_G_H	ETS(M,A,M)	ETS	186K	5.9%	C_I_G	Nixtla_AutoETS	Nixtla	72K	2.4%
A_F_L	Nixtla_OptTheta	Nixtla	161K	7.4%	B_I_K	ETS(A,Md,A)	ETS	274K	5.4%
A_G_F	SeasonalNaive	Base	27K	2.0%	C_I_K	Nixtla_AutoARIMA	Nixtla	33K	2.2%
C_G_H	Nixtla_OptTheta	Nixtla	25K	2.8%	A_A_E	SeasonalNaive	Base	39K	2.8%

The table above shows the top-ranked series by CV RMSE. Notably, the best models span *all four families*—ETS variants (multiplicative, additive, damped trends), Nixtla automated models (OptTheta, AutoARIMA, AutoETS), and even baselines for the most stable series. This diversity confirms that no single model architecture is universally superior and validates the per-series selection approach.

Reliability Analysis

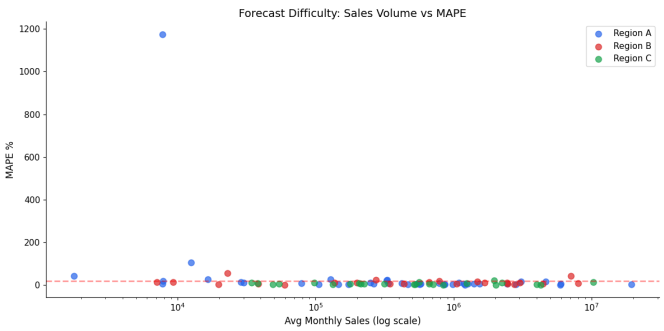

Figure 7: MAPE vs. avg sales volume—low-volume series drive the highest forecast errors.

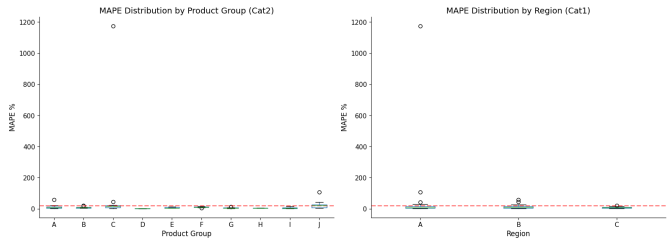

Figure 8: MAPE distribution by product group and region. Group J and Region A show widest spread.

Forecast accuracy strongly correlates with sales volume (Figure 7): series above 100K avg monthly sales consistently achieve MAPE under 10%, while the unreliable forecasts concentrate in low-volume series (<10K). This is inherent to the data—small volumes have high coefficients of variation where a difference of a few hundred units translates to large percentage errors. Three specific patterns drive high MAPE: Product Group J (median 22%, erratic non-seasonal demand), NULL sub-categories (median 16%, lack of product specificity), and low-volume series (<10K avg, median 35%). These are **data-inherent limitations**, not modeling failures—no model family could reliably predict these series.

Conclusion

This project demonstrates that a **well-structured multi-model framework** can deliver reliable forecasts even with limited history (44 months) and messy real-world data. The key insight is that *no single model dominates*—ETS wins 41% of series, Nixtla 34%, and even simple baselines outperform complex methods on 18% of series where noise exceeds signal. This heterogeneity justifies the per-series selection approach over a one-size-fits-all strategy.

The **87% of series with MAPE below 20%** represents a strong forecast foundation for business decision-making. The combination of classical statistical methods (ETS, SARIMAX), automated Nixtla models, and ML provided robust coverage across diverse series dynamics. Key methodological decisions—AIC for model selection (appropriate for short-series forecasting per Hyndman & Athanasopoulos), individual series treatment (avoiding signal contamination from aggregation), and expanding-window CV—were grounded in the constraints of short-history, high-seasonality sales data.

The remaining 13% of series with poor forecasts are not modeling failures but reflect **inherent data limitations**: low-volume products with erratic demand patterns, missing sub-category specificity (NULL groups), and product groups with non-seasonal dynamics (Group J). For these series, the recommendation is to flag them for qualitative business judgment and supplement algorithmic forecasts with domain expertise.

Strengths & Limitations

Strengths:

- 14 model types across 4 families ensure broad coverage
- Per-series selection via CV—no one-size-fits-all assumption
- Principled AIC selection for SARIMAX, appropriate for short series
- Full portfolio: all 87 viable series forecasted with documented metrics
- Diversity-based ensemble as safety net against model failure
- Transparent quality tiers for downstream business decisions

Limitations & Next Steps:

- No exogenous variables (promotions, economic indicators)—SARIMAX supports them if available
- No hierarchical reconciliation—MinT optimal would ensure sub-product forecasts sum correctly
- Point forecasts only—prediction intervals via bootstrapping or conformal prediction needed
- ML limited by per-series data scarcity—global models across all series could share patterns
- No VAR: 87 series $\times$ k^2 parameters/lag = dimensionality explosion; correlation $\neq$ bidirectional causation